

FACE RECOGNITION Using Raspberry Pi

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To secure our assets and protect our privacy, there is a strong demand for user-friendly biometric security systems. There are various types of biometric systems that use signatures, fingerprints, voice, hand geometry, face geometry, face detection and so on. Among these, face recognition appears to be quite exciting and is catching attention.

Circuit and working

This project requires Raspberry Pi B+, Raspberry Pi camera, a pushbutton switch, a relay and some miscellaneous components.

Face images are captured through Raspberry Pi camera and stored in a database in Raspberry Pi. To capture your face image, place yourself in front of the Pi camera and press pushbutton switch S1. The image of your face will get stored in the database. Next time when you face Pi camera and press S1, your face will be recognised, relay RL1 will be energised and your electrical load/solenoid will be activated.

Software

This project uses OpenCV computer vision library to perform face detection and recognition.

First, install OpenCV dependencies. Compiling OpenCV on Raspberry Pi may take about five hours (depending on your system and network speed). So make sure

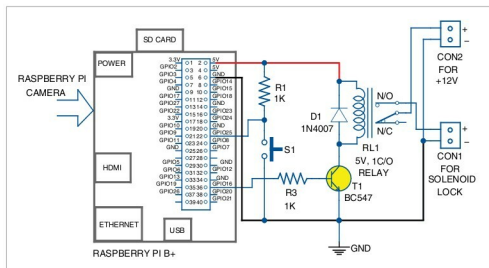


Fig. 1: Circuit diagram of the face-recognition system using Raspberry Pi

you have sufficient time to start the process before proceeding.

Power on Raspberry Pi, open the terminal, set up Wi-Fi and execute the following commands:

```
$ sudo apt-get update
$ sudo apt-get upgrade
$ sudo apt-get install build-essential
cmake pkg-config python-dev libgtk2.0-dev
libgtk2.0-dev libpng-dev libjpeg-dev
libtiff-dev libjasper-dev
libavcodec-dev swig unzip
```

Select yes for all options and wait for the libraries and dependencies to be installed.

Now, unzip OpenCV directory by executing the following commands:

```
$ wget http://downloads.sourceforge.net/
project/opencvlibrary/opencv-unix/2.4.9/
opencv-2.4.9.zip
$ unzip opencv-2.4.9.zip
```

Change the directory and execute cmake command as given below to build the makefile:

```
$ cd opencv-2.4.9
```

```
$ cmake -DCMAKE_BUILD_TYPE=RELEASE
-DCMAKE_INSTALL_PREFIX=/usr/local
-DBUILD_PERF_TESTS=OFF -DBUILD_opencv_
gpu=OFF -DBUILD_opencv_ocl=OFF
```

Compile the project by executing the command given below:

```
$ make
```

It may take about five hours for compilation.

Install the compiled OpenCV libraries by executing the following command:

```
$ sudo make install
```

The latest version of OpenCV is now installed on your Raspberry Pi.

Face-recognition code is written in Python, so some dependencies have to be installed using the following commands:

```
$ sudo apt-get install python-pip
$ sudo apt-get install python-dev
$ sudo pip install picamera
$ sudo pip install rpio
```

After OpenCV and Python dependencies are installed, the project